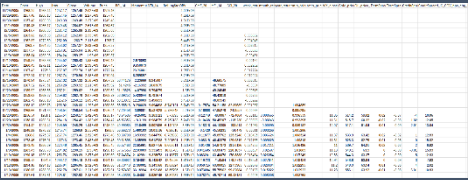


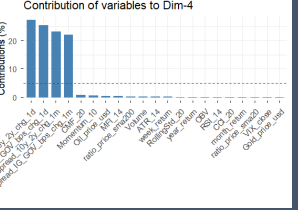
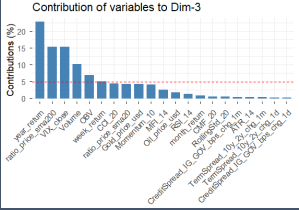
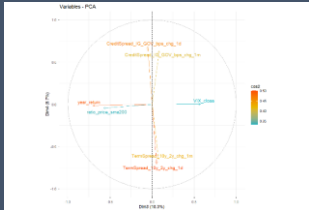
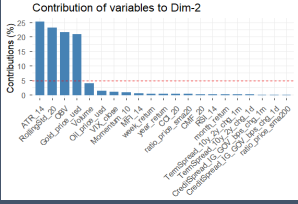
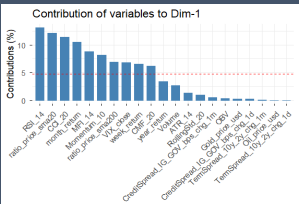
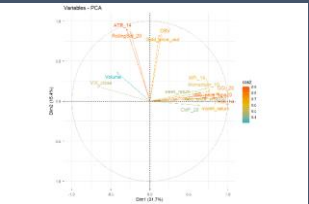
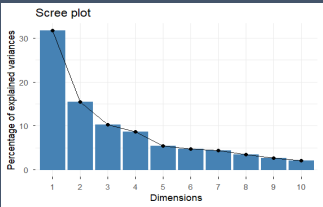
Projet en analyse de données

Étant étudiants en finance quantitative, nous nous demandions s’il serait possible de prédire si le rendement du S&P 500. L’objectif serait d’obtenir un rendement supérieur au S&P 500 à long terme en retirant notre investissement lors d’une prédiction d’un rendement négatif. Afin d’y arriver nous avons fait une analyse en composante principal pour choisir les meilleurs indicateurs financiers. Ensuite, nous avons entraîné un algorithme XGBoost avec les indicateurs choisis afin d’essayer de prédire le signe du rendement de chaque période de 21 jours des 10 dernières années.

Données initiales : Rendements du S&P 500 et de différents indicateurs techniques et macroéconomiques.



Résultats de l’analyse en composante principale:



Corrélations des indicateurs afin d’éviter les indicateurs similaires

RSI_14	1
CCI_20	0.901523905
ratio_price_sma20	0.840816293
month_return	0.779922121
MFI_14	0.766933621
Momentum_10	0.689701095
CMF_20	0.649107213
week_return	0.624668705
ratio_price_sma200	0.553335165
year_return	0.337322162
OBV	0.127174425
Gold_price_usd	0.106241596
TermSpread_10y_2y_chg_1m	0.006143995
TermSpread_10y_2y_chg_1d	0.005098144
Oil_price_usd	-0.056417831
CreditSpread_IG_GOV_bps_chg_1d	-0.1064328
RollingStd_20	-0.162551677
CreditSpread_IG_GOV_bps_chg_1m	-0.168712627
ATR_14	-0.236251952
Volume	-0.256909557
VIX_close	-0.510602874

ATR_14	1
RollingStd_20	0.894944235
OBV	0.663748833
Gold_price_usd	0.632209755

OBV	1
Gold_price_usd	0.776085594

year_return	1
ratio_price_sma200	0.755180827
OBV	0.342097077
RSI_14	0.337322162
Gold_price_usd	0.293352358
month_return	0.255191284
MFI_14	0.231419038
CCI_20	0.231302622
ratio_price_sma20	0.220162426
CMF_20	0.20394557
Momentum_10	0.187151735
week_return	0.132673961
Oil_price_usd	0.100425641
TermSpread_10y_2y_chg_1d	-0.005667207
TermSpread_10y_2y_chg_1m	-0.027997578
CreditSpread_IG_GOV_bps_chg_1d	-0.029076348
CreditSpread_IG_GOV_bps_chg_1m	-0.077131698
RollingStd_20	-0.086027496
ATR_14	-0.124402847
Volume	-0.3663308
VIX_close	-0.598673748

TermSpread_10y_2y_chg_1d	1	0.245208516	-0.509123466	-0.109136656
TermSpread_10y_2y_chg_1m	0.245208516	1	-0.096249749	-0.437026814
CreditSpread_IG_GOV_bps_chg_1d	-0.509123466	-0.096249749	1	0.234586204
CreditSpread_IG_GOV_bps_chg_1m	-0.109136656	-0.437026814	0.234586204	1

Prédictions des Rendements du S&P 500

Date	Volume	RSI_14	CCI_20	ratio_price_sma20	month_return	MFI_14	Momentum_10	CMF_20	week_return	ratio_price_sma200	year_return	OBV	Gold_price_usd	TermSpread_10y_2y_chg_1d	CreditSpread_IG_GOV_bps_chg_1d	RollingStd_20	CreditSpread_IG_GOV_bps_chg_1m	ATR_14	Volume	VIX_close
2008-01-02	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-01-09	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-01-16	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-01-23	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-01-30	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-02-06	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-02-13	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-02-20	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-02-27	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-03-06	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-03-13	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-03-20	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-03-27	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-04-03	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-04-10	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-04-17	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-04-24	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-05-01	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-05-08	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-05-15	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-05-22	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-05-29	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-06-05	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-06-12	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-06-19	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-06-26	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-07-03	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-07-10	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-07-17	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-07-24	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-07-31	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-08-07	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-08-14	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-08-21	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-08-28	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-09-04	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-09-11	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-09-18	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-09-25	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-10-02	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-10-09	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-10-16	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-10-23	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-10-30	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-11-06	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-11-13	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-11-20	1000000000	50.0	100.0	1.0	0.0	50.0	100.0	0.0	0.0	1.0	0.0	0.0	1000.0	0.0	0.0	0.0	0.0	0.0	1000000000	1.0
2008-11-27	1																			